

homework_2

Question 1

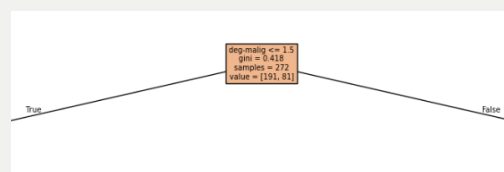
Learn a Decision Tree from the whole dataset by setting the max depth threshold to 5, while keeping the default configuration for all the other parameters.

```
# Initialize the Decision Tree Classifier, set max depth
clf = DecisionTreeClassifier(max_depth=5)
# Train the Decision Tree Classifier
clf.fit(X, y)
```

Point A

Which attribute is deemed to be the most discriminative one for class prediction?

The most discriminative attribute in this Decision Tree would be the one selected as the root node, since it provides the best split of the dataset according to the information gain or Gini impurity criterion. In our case, this feature is "deg_malig".



Point B

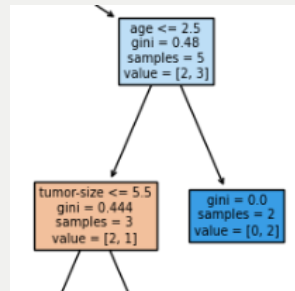
What is the height of the Decision Tree generated?

The height of the Decision Tree is 5, which corresponds to the max_depth parameter we set. This means the tree has 5 levels of nodes from the root to the deepest leaf nodes. The height represents the longest path from the root to any leaf node in our decision tree structure.

Point C

Find a pure partition in the Decision Tree and report a screenshot that shows the example identified.

A pure partition in our Decision Tree refers to a node where all samples belong to the same class.



Question 2

Analyze the impact of the minimal impurity (using the entropy splitting criterion), minimum number of samples to split, minimum number of samples for each leaf and maximal depth parameters on the characteristics on the Decision Tree model learnt from the whole dataset (keep the default configuration for all the other parameters). Report at least 5 different screenshots showing Decision Trees (or portions of them) generated with different configuration settings.

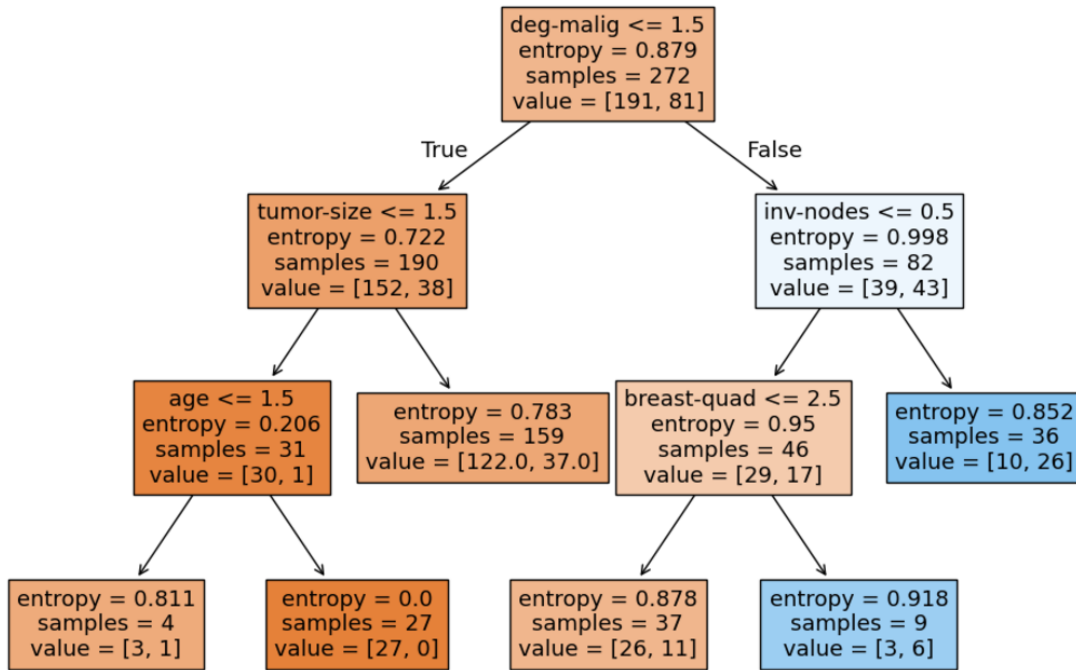
```
parameters = [
    {"min_samples_split": 3, "min_samples_leaf": 3, "max_depth": 3, "min_impurity_decrea
    {"min_samples_split": 2, "min_samples_leaf": 4, "max_depth": 3, "min_impurity_decrea
    {"min_samples_split": 3, "min_samples_leaf": 3, "max_depth": 4, "min_impurity_decrea
    {"min_samples_split": 2, "min_samples_leaf": 2, "max_depth": 4, "min_impurity_decrea
    {"min_samples_split": 2, "min_samples_leaf": 3, "max_depth": 3, "min_impurity_decrea
]

for param in parameters:
    clf = DecisionTreeClassifier(
        criterion='entropy',
        min_samples_split=param["min_samples_split"],
        min_samples_leaf=param["min_samples_leaf"],
        max_depth=param["max_depth"],
        min_impurity_decrease=param["min_impurity_decrease"],
    )

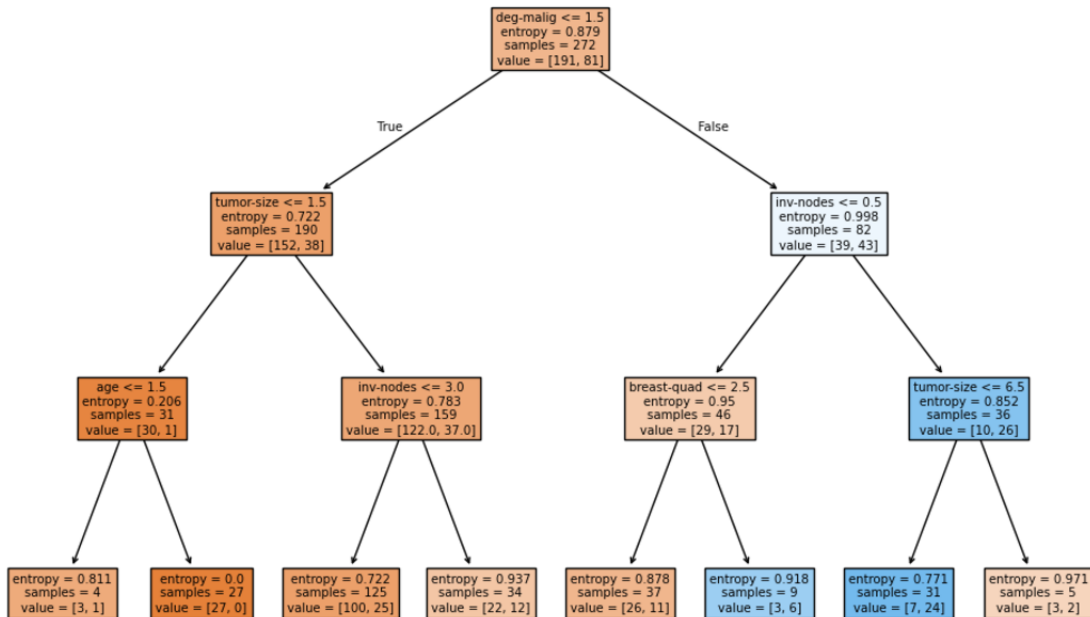
    clf.fit(X, y)

    plt.figure(figsize=(12, 8))
    plot_tree(clf, feature_names=X.columns, filled=True)
    plt.title(f"Tree with params: {param}")
    plt.show()
```

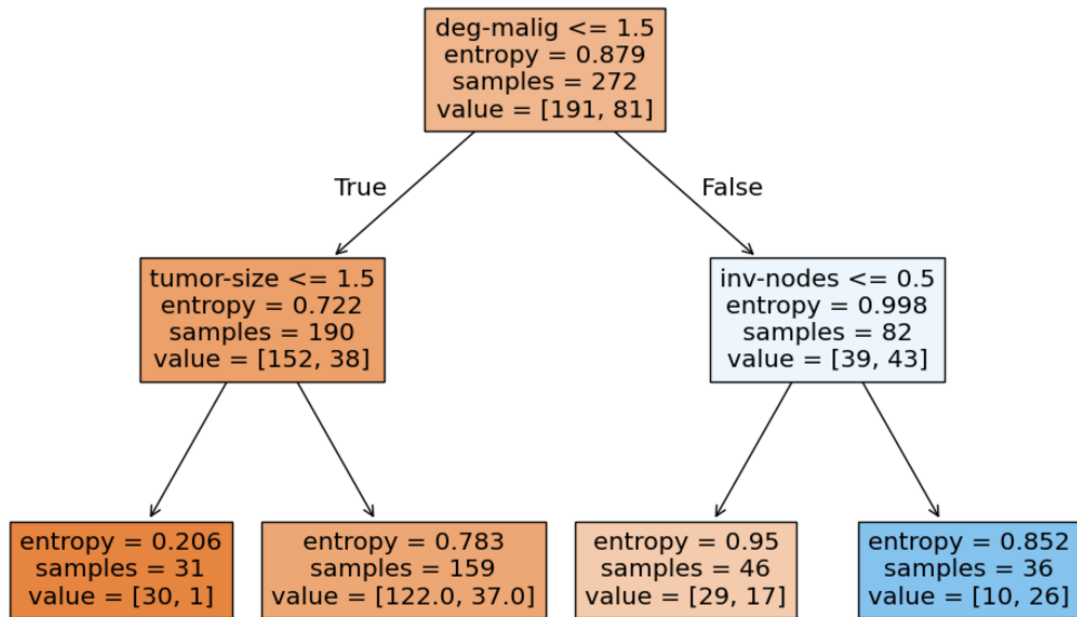
Tree with params: {'min_samples_split': 3, 'min_samples_leaf': 3, 'max_depth': 3, 'min_impurity_decrease': 0.01}



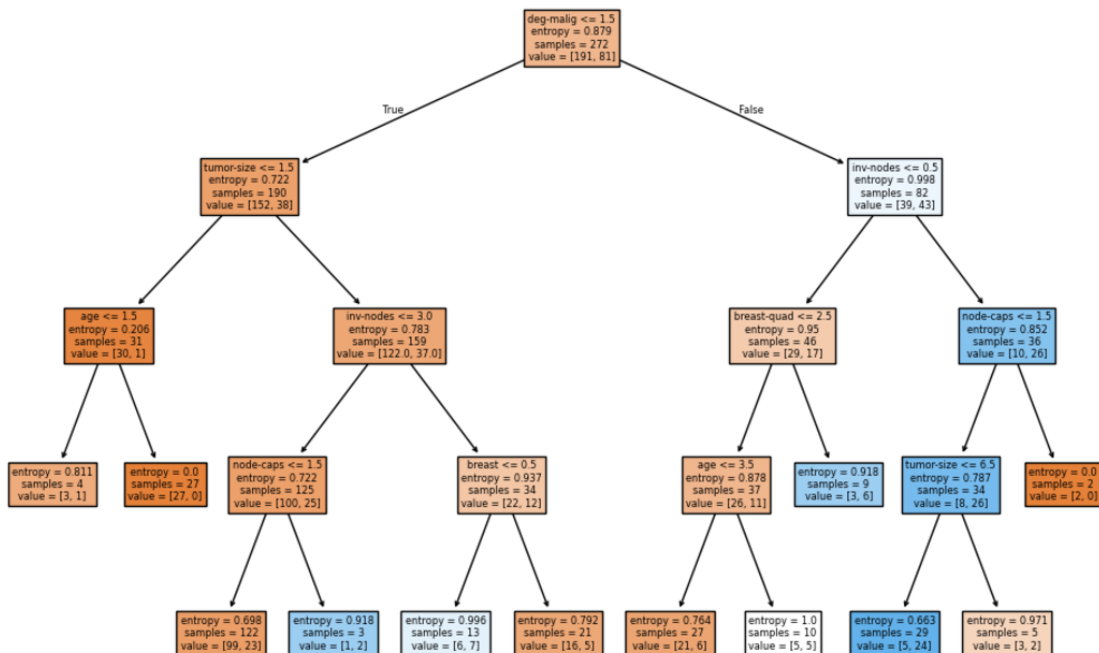
Tree with params: {'min_samples_split': 2, 'min_samples_leaf': 4, 'max_depth': 3, 'min_impurity_decrease': 0.001}



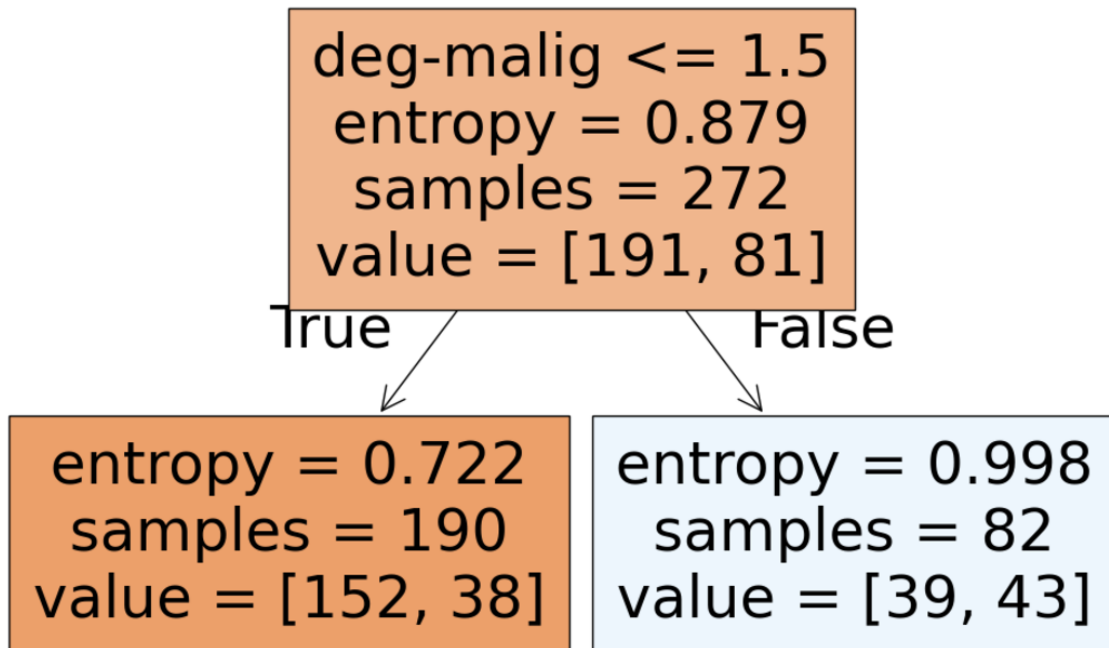
Tree with params: {'min_samples_split': 3, 'min_samples_leaf': 3, 'max_depth': 4, 'min_impurity_decrease': 0.02}



Tree with params: {'min_samples_split': 2, 'min_samples_leaf': 2, 'max_depth': 4, 'min_impurity_decrease': 0.005}



Tree with params: {'min_samples_split': 2, 'min_samples_leaf': 3, 'max_depth': 3, 'min_impurity_decrease': 0.05}



Question 3

Performing a 10-fold Stratified Cross-Validation, what is the impact the minimal impurity, minimum number of samples to split, minimum number of samples for each leaf and maximal depth parameters on the average accuracy achieved by Decision Tree?

```
parameters = [
    {"min_samples_split": 3, "min_samples_leaf": 3, "max_depth": 3, "min_impurity_decrea
    {"min_samples_split": 2, "min_samples_leaf": 4, "max_depth": 3, "min_impurity_decrea
    {"min_samples_split": 3, "min_samples_leaf": 3, "max_depth": 4, "min_impurity_decrea
    {"min_samples_split": 2, "min_samples_leaf": 2, "max_depth": 4, "min_impurity_decrea
    {"min_samples_split": 2, "min_samples_leaf": 3, "max_depth": 3, "min_impurity_decrea
]

skf = StratifiedKFold(n_splits=10, shuffle=True)

for param in parameters:
    clf = DecisionTreeClassifier(
        criterion='entropy',
        min_samples_split=param["min_samples_split"],
        min_samples_leaf=param["min_samples_leaf"],
        max_depth=param["max_depth"],
        min_impurity_decrease=param["min_impurity_decrease"],
    )

    scores = cross_val_score(clf, X, y, cv=skf)
```

```
print(f"Average accuracy for params {param}:\n{scores.mean()}")

y_pred = cross_val_predict(clf, X, y, cv=skf)
cm = confusion_matrix(y, y_pred)
print(f"Confusion matrix for params {param}:")
print(cm)
print()
```

The third model achieves higher accuracy due to a **maximum depth of 4** (`max_depth`). Additionally, the `min_samples_leaf=3` parameter ensures a good balance between generalization and fitting to the data. This combination of parameters creates a tree that is sufficiently complex to model the data but well-regulated to avoid errors.

```
Average accuracy for params {'min_samples_split': 3, 'min_samples_leaf': 3, 'max_depth': 3, 'min_impurity_decrease': 0.01}:
0.7203703703703705
Confusion matrix for params {'min_samples_split': 3, 'min_samples_leaf': 3, 'max_depth': 3, 'min_impurity_decrease': 0.01}:
[[173  18]
 [ 57  24]]

Average accuracy for params {'min_samples_split': 2, 'min_samples_leaf': 4, 'max_depth': 3, 'min_impurity_decrease': 0.001}:
0.7133597883597884
Confusion matrix for params {'min_samples_split': 2, 'min_samples_leaf': 4, 'max_depth': 3, 'min_impurity_decrease': 0.001}:
[[174  17]
 [ 64  17]]

Average accuracy for params {'min_samples_split': 3, 'min_samples_leaf': 3, 'max_depth': 4, 'min_impurity_decrease': 0.02}:
0.7538359788359787
Confusion matrix for params {'min_samples_split': 3, 'min_samples_leaf': 3, 'max_depth': 4, 'min_impurity_decrease': 0.02}:
[[181  10]
 [ 58  23]]

Average accuracy for params {'min_samples_split': 2, 'min_samples_leaf': 2, 'max_depth': 4, 'min_impurity_decrease': 0.005}:
0.7132275132275132
Confusion matrix for params {'min_samples_split': 2, 'min_samples_leaf': 2, 'max_depth': 4, 'min_impurity_decrease': 0.005}:
[[165  26]
 [ 60  21]]

Average accuracy for params {'min_samples_split': 2, 'min_samples_leaf': 3, 'max_depth': 3, 'min_impurity_decrease': 0.05}:
0.6911375661375662
Confusion matrix for params {'min_samples_split': 2, 'min_samples_leaf': 3, 'max_depth': 3, 'min_impurity_decrease': 0.05}:
[[153  38]
 [ 42  39]]
```

Question 4

Considering the K-Nearest Neighbor (K-NN) classifier and performing a 10-fold Stratified CrossValidation, what is the impact of parameter K on the average classifier accuracy? Report at least 5 screenshots showing the confusion matrices achieved using different K parameter values.

```
# Initialize the KNN classifier
neighbors = [3, 4, 5, 6, 7]

# Perform StandardScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Perform cross-validation predictions
skf = StratifiedKFold(n_splits=10)
for n in neighbors:
    clf = KNeighborsClassifier(n_neighbors=n)
```

```

scores = cross_val_score(clf, X_scaled, y, cv=skf)

print(f"KNN Neighbors: {n}")

# Confusion matrix
y_pred = cross_val_predict(clf, X_scaled, y, cv=skf)
cm = confusion_matrix(y, y_pred)
print("Confusion Matrix:")
print(cm)

# Evaluate accuracy
print(f"Accuracy: {scores.mean()}")
print()

```

The mean accuracy tends to increase initially with the increase of k, reaching a peak at K=4 and K=5, but with higher values tends to decrease lightly.

```

KNN Neighbors: 3
Confusion Matrix:
[[165  26]
 [ 56  25]]
Accuracy: 0.6985449735449736

KNN Neighbors: 4
Confusion Matrix:
[[178  13]
 [ 65  16]]
Accuracy: 0.7134920634920635

KNN Neighbors: 5
Confusion Matrix:
[[171  20]
 [ 58  23]]
Accuracy: 0.7134920634920635

KNN Neighbors: 6
Confusion Matrix:
[[177  14]
 [ 65  16]]
Accuracy: 0.7099206349206348

KNN Neighbors: 7
Confusion Matrix:
[[170  21]
 [ 61  20]]
Accuracy: 0.6986772486772488

```

Perform a 10-fold Stratified Cross-Validation with classifier Naïve Bayes. Does K-NN perform on average better or worse than the Naïve Bayes classifier on the analyzed data? Report a screenshot showing the confusion matrix achieved by Naïve Bayes on the analyzed dataset.

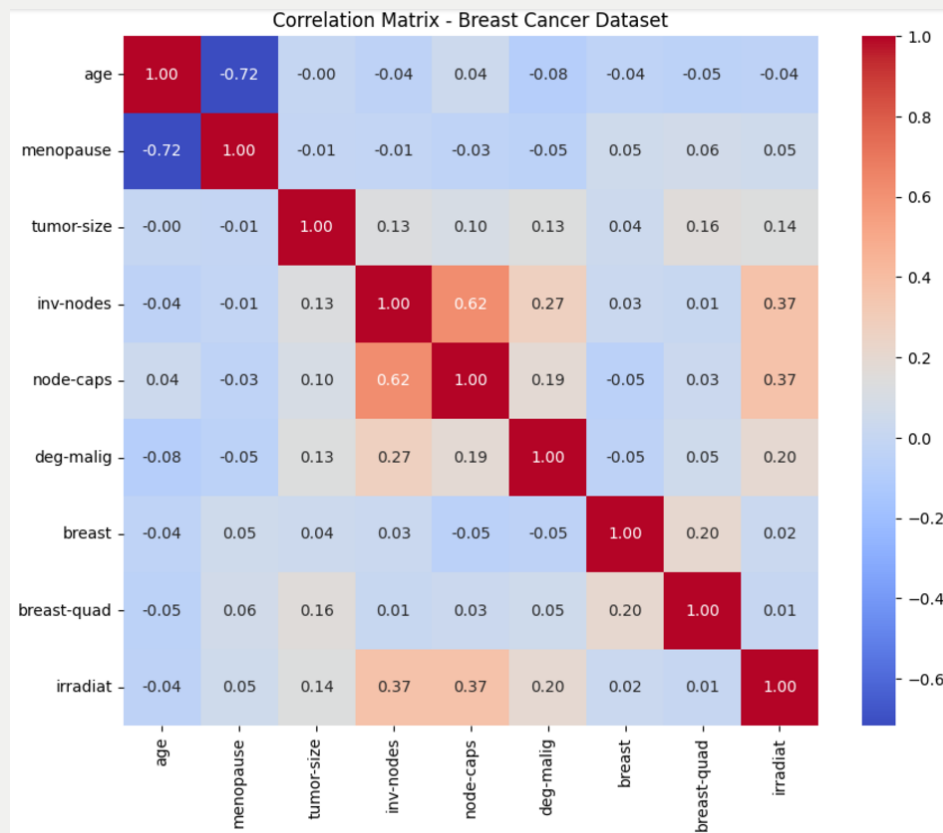
Based on the confusion matrix obtained from the Naive Bayes on the analyzed dataset, it is possible to say that the mean accuracy is better as compared to KNN obtained with the previous configurations, reported above.

```
Confusion Matrix:  
[[157  34]  
 [ 41  40]]  
Accuracy: 0.725
```

Question 5

Analyze the Correlation Matrix to discover pairwise correlations between data attributes. Report a screenshot showing the correlation matrix achieved.

```
# Initialize the Naive Bayes classifier  
clf = GaussianNB()  
  
# Perform cross-validation predictions  
skf = StratifiedKFold(n_splits=10)  
scores = cross_val_score(clf, X, y, cv=skf)  
  
# Confusion matrix  
y_pred = cross_val_predict(clf, X, y, cv=skf)  
cm = confusion_matrix(y, y_pred)  
print("Confusion Matrix:")  
print(cm)  
  
# Evaluate accuracy  
print(f"Accuracy: {scores.mean()}")  
print()
```

Point A

Does the Naïve independence assumption actually hold for the Breast dataset?

No, the Naive independence assumption is not valid for this dataset, since not all attributes are uncorrelated or with a degree of correlation under a certain threshold .

Point B

Which is the pair of most correlated attributes?

The most correlated pair of attributes is the one containing `menopause` and `age` attributes, which have a strong negative correlation. This shows that when a value of an attribute increases, the one from the other attribute decreases in a significant way.